PV102 Day 4 Student Packet

Module Behavior String Design and Shade

Today's rule

String design follows limits, temperature, and shade reality.

Learning targets

- Predict how series and parallel wiring affect PV string voltage and current.
- Explain why controller input limits control string design.
- Recognize shade as both a production and mismatch problem.
- Use a safe paper/string model to compare design options before hardware work.

Morning quiz

1. In series wiring, matching module voltage primarily:

- A. Adds
- B. Disappears
- C. Becomes AC
- D. Turns into resistance

2. In parallel wiring, matching branch current primarily:

- A. Adds
- B. Becomes zero
- C. Changes to frequency
- D. Removes voltage

3. The charge controller input limit is important because:

- A. Exceeding limits can damage equipment or create unsafe conditions
- B. It chooses wire color
- C. It makes sunlight brighter
- D. It stores energy

4. Shade on one module can affect:

- A. Only the paper worksheet
- B. Production and string behavior
- C. Lunch time only
- D. Nothing if the roof is new

5. Cold temperature generally makes PV voltage:

- A. Higher
- B. Always zero
- C. AC only
- D. Unrelated to design

6. Why should string design be checked before connecting equipment?

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7. A controller allows 100 V max input. Should a design near 99 V be accepted without temperature check?

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8. Name one reason parallel strings may require added protection.

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9. What evidence from Day 2 matters most for shade design?

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10. Write the Day 4 rule in your own words.

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Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

String Design Paper Lab

Teams compare three paper string options against controller limits and shade constraints.

Scenario options

- Four 22 Vmp / 27 Voc training modules feeding a 100 V controller.
- Two shaded morning modules and two unshaded modules on a shed roof.
- Long ground-rack wire run where voltage drop and current both matter.

Team procedure

- Read the module and controller cards.
- Build three possible string layouts using cards.
- Calculate approximate operating voltage/current and open-circuit voltage.
- Mark shade exposure from Day 2 scenario notes.
- Reject any option that violates a limit or ignores shade evidence.
- Handoff the safest first-pass string layout and its required verification checks.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

What made a string option unacceptable today: equipment limit, shade reality, or missing information?

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